

Recurrence relations

(from Paragraphs 8.1-8.3)

Fast multiplication algorithm (p.528)

Recall that multiplying two binary numbers takes $O(n^2)$ time.
Sum of two binary numbers takes $O(n)$.

The Karatsuba algorithm is a fast multiplication algorithm.

It was discovered by Anatoly Karatsuba in 1960 and published in 1962.

It reduces the multiplication of two n -digit numbers to $O(n^{1.58})$.

For example, the Karatsuba algorithm requires 59,049 single-digit multiplications to multiply two 1024-digit numbers, whereas the classical algorithm requires 1,048,576 (a speedup of 17.75 times).

Let $a = (a_{2n-1} \dots a_1 a_0)_2$ and $b = (b_{2n-1} \dots b_1 b_0)_2$.

Denote

$$A_1 = (\overbrace{a_{2n-1} \dots a_n}^n)_2, \quad A_0 = (\overbrace{a_{n-1} \dots a_0}^n)_2$$

$$B_1 = (b_{2n-1} \dots b_n)_2, \quad B_0 = (b_{n-1} \dots b_0)_2.$$

$$a = A_0 + 2^n A_1, \quad b = B_0 + 2^n B_1,$$

$$ab = (2^{2n} + 2^n) A_1 B_1 + 2^n (A_1 - A_0)(B_1 - B_0) + (2^n + 1) A_0 B_0$$

$$T(2n) = 3T(n) + Cn$$

or

$$T(n) = 3T\left(\frac{n}{2}\right) + Cn$$

By Master theorem $a=3, b=2, d=1, 3 > 2^1$
hence $O(n^{\log_2 3 \approx 1.58})$

Answer Q1

Fibonacci numbers

The **Fibonacci numbers**, $f_0, f_1, f_2, \dots$, are defined by $f_0 = 0$, $f_1 = 1$, and

$$\underline{f_n = f_{n-1} + f_{n-2}}, \quad n = 2, 3, 4, \dots$$

The sequence does have lots of other interesting and practical applications.

Please read more about it [here.](#)

The original problem that Fibonacci investigated (in the year 1202) was about how fast rabbits could breed in ideal circumstances.

Suppose a newly-born pair of rabbits, one male, one female, are put in a field. At the end of its second month a female can produce another pair of rabbits. Suppose that our rabbits never die and that the female always produces one new pair (one male, one female) every month from the second month on. The puzzle that Fibonacci posed was How many pairs will there be in one year?

Reproducing pairs (at least two months old)	Young pairs (less than two months old)	Month	Total pairs
		1	1
		2	1
		3	2
		4	3
		5	5
		6	8

The number of pairs of rabbits in the field at the start of each month is 1, 1, 2, 3, 5, 8, 13, 21, 34, ...

Answer Q2

Algorithms of finding Fibonacci Numbers

Recursive Algorithm

```
procedure fibonacci( $n$ : nonnegative integer)
if  $n = 0$  then return 1
else if  $n = 1$  then return 1
else return fibonacci( $n - 1$ ) + fibonacci( $n - 2$ )
```

Handwritten notes and diagram:
Sequence: (0), 1, 1, (2), 3, 5, 8, ...
Annotations: $f_0 = 1$, $f_1 = 1$, $f_2 \leftarrow f_1 + f_0$, $f_3 \leftarrow f_2 + f_1$, $f_4 \leftarrow f_3 + f_2$, $f_5 \leftarrow f_4 + f_3$.
Diagram shows arrows indicating the recursive calls for f_3 and f_4 . f_3 is circled in red.

Number of additions is $f_n - 1$, this is $O(2^n)$.

Iterative Algorithm

```
procedure iterative_fibonacci( $n$ : nonnegative integer)
if  $n = 0$  then return 1
else  $x := 1, y := 1$ 
    for  $i := 1$  to  $n - 1$ 
         $z := x + y$ 
         $x := y$ 
         $y := z$ 
    return  $y$ 
```

Handwritten note: f_0 with an arrow pointing to the $n = 0$ case.

Number of additions is $O(n)$.

Show that whenever $n \geq 3$,

$$f_n > \left(\frac{1 + \sqrt{5}}{2} \right)^{n-2} > \left(\frac{3}{2} \right)^n$$

Handwritten notes: "2..." above the fraction, and "2" below the fraction.

Solution:

$$P(n) = "f_n > 2^{n-2}"$$

$$2 < 2 = f$$

$$B.S.: f_3 > 2 \Rightarrow P(3) \equiv 1$$

$$I.S.: f_j > 2^{j-2}, \quad 3 \leq j \leq k$$

$$2 \text{ is solution of } d^2 - d - 1 = 0 \Rightarrow d^2 = d + 1 \text{ (check!)}$$

$$d^{k-1} = d^2 d^{k-3} = (d+1) d^{k-3} = d^{k-2} + d^{k-3}$$

$$f_{k+1} = f_k + f_{k-1} > d^{k-2} + d^{k-3} \leftarrow = d^{k-1}$$

$$\Rightarrow P(k+1) \equiv 1 \text{ By S.I. } \forall n P(n) \equiv 1 \quad \square$$

Euclidean algorithm uses $O(\log b)$

Lamé's theorem

Let a and b be positive integers with $a \geq b$. Then the number of divisions used by the Euclidean algorithm to find $\gcd(a, b)$ is less than or equal to five times the number of decimal digits in b . ($\log b$)

Proof: To find $\gcd(a, b)$, $a \geq b$, $r_0 = a$, $r_1 = b$

$$\begin{cases} r_0 = r_1 q_1 + r_2 & 0 \leq r_2 < r_1 \\ r_1 = r_2 q_2 + r_3 & 0 \leq r_3 < r_2 \\ \vdots \\ r_{n-2} = r_{n-1} q_{n-1} + r_n, & 0 \leq r_n < r_{n-1} \\ r_{n-1} = r_n q_n + 0 & \Rightarrow \gcd(a, b) = r_n \end{cases}$$

$q_n \geq 2$, because $(r_n < r_{n-1})$

$\Rightarrow r_n \geq 1 = f_2$, $r_{n-1} \geq 2r_n \geq 2f_2 = f_3$

$$r_n \geq 1 = f_2, \quad r_{n-1} \geq 2f_2 = f_3$$

$$r_{n-2} \geq r_{n-1} + r_n \geq f_2 + f_3 = f_4$$

...

$$r_2 \geq r_3 + r_4 \geq f_{n-1} + f_{n-2} = f_n$$

$$b = r_1 \geq r_2 + r_3 \geq f_n + f_{n-1} = f_{n+1}$$

$$\Rightarrow b \geq f_{n+1}, \text{ we also have } f_{n+1} > 2^{n-1}$$

$$\Rightarrow b > 2^{n-1}$$

$$\log_{10} b > (n-1) \log_{10} 2 > \frac{n-1}{5} \left(\log_{10} 2 > \frac{1}{5} \right)$$

$$\Rightarrow n-1 < 5 \cdot \log_{10} b \Rightarrow n < 1 + 5 \cdot \log_{10} b$$

$$n = O(\log b)$$



$$\alpha = \frac{1+\sqrt{5}}{2}$$

" $n-1$ "

Solving linear recurrence relations

(Paragraph 8.2)

Linear homogeneous recurrence relations

A **linear homogeneous recurrence relation of degree k with constant coefficients** is a recurrence relation of the form

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-\underline{k}},$$

where $c_1, c_2, \dots, c_k$ are real numbers, and $c_k \neq 0$.

A consequence of the principle of mathematical induction is that a sequence satisfying the recurrence relation in the definition is **uniquely determined** by this recurrence relation and the k initial conditions

$$a_0 = C_0, \quad a_1 = C_1, \dots, \quad a_{k-1} = C_{k-1}.$$

$P_n = 1.11 \cdot P_{n-1}$ is a lin. hom. rec. rel. of deg. 1 with const. coeff.
 $f_n = f_{n-1} + f_{n-2}$ ——— || ——— of deg 2 — || ———
 $a_n = a_{n-5}$ ——— || ——— of deg 5 — || ———
 $a_n = a_{n-1} + a_{n-2}^2$ not linear
 $H_n = 2H_{n-1} + 1$ not homogeneous.
 $B_n = nB_{n-1}$ does not have constant coeff.

Answer Q3

Theorem 1

Suppose that $x^2 - c_1x - c_2 = 0$ has two distinct roots x_1 and x_2 .

Then the sequence $\{a_n\}$ is a solution of the recurrence relation

$a_n = c_1a_{n-1} + c_2a_{n-2}$ if and only if

$$a_n = \alpha_1 x_1^n + \alpha_2 x_2^n$$

for $n = 0, 1, 2, \dots$, where α_1 and α_2 are constants. The equation $x^2 - c_1x - c_2 = 0$ is called the characteristic equation and x_1 and x_2 are characteristic roots.

Solve the recurrence relation $a_n^{x^2} = a_{n-1}^{x'} + 2a_{n-2}^{x^0}$ with $a_0 = 2$ and $a_1 = 7$.

Solution:

$$x^2 - x - 2 = 0 \Rightarrow x_1 = 2, x_2 = -1$$

$\Rightarrow a_n = d_1 2^n + d_2 (-1)^n$ to find d_1 and d_2 use init. cond.

$$\begin{cases} 2 = d_1 + d_2 \\ 7 = 2d_1 + d_2 \end{cases} \Rightarrow 3d_1 = 9, \Rightarrow d_1 = 3, \Rightarrow d_2 = -1 \Rightarrow \boxed{a_n = 3 \cdot 2^n - (-1)^n}$$

Fibonacci numbers

Recall that the sequence of Fibonacci numbers satisfies the recurrence relation $f_n = f_{n-1} + f_{n-2}$ and also satisfies the initial conditions $f_0 = 0$ and $f_1 = 1$.

char. eq. $x^2 - x - 1 = 0 \Rightarrow x_1 = \frac{1+\sqrt{5}}{2}, x_2 = \frac{1-\sqrt{5}}{2}$

$$f_n = d_1 \left(\frac{1+\sqrt{5}}{2} \right)^n + d_2 \left(\frac{1-\sqrt{5}}{2} \right)^n$$

$$\begin{cases} 0 = d_1 + d_2 \end{cases}$$

$$\begin{cases} 1 = d_1 \frac{1+\sqrt{5}}{2} + d_2 \frac{1-\sqrt{5}}{2} \Rightarrow 2 = d_1 + d_2 + \sqrt{5}(d_1 - d_2) \Rightarrow \end{cases}$$

$$\Rightarrow \frac{2}{\sqrt{5}} = d_1 - d_2 \Rightarrow d_1 = \frac{1}{\sqrt{5}}, d_2 = -\frac{1}{\sqrt{5}}$$

$$\Rightarrow \boxed{f_n = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^n - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^n}$$

Theorem 2

Suppose that $x^2 - c_1x - c_2 = 0$ has only one root x_0 . $= (x - x_0)^2$

Then the sequence $\{a_n\}$ is a solution of the recurrence relation

$a_n = c_1a_{n-1} + c_2a_{n-2}$ if and only if

$$a_n = \alpha_1 x_0^n + \alpha_2 n x_0^n = (d_1 + d_2 n) x_0^n$$

for $n = 0, 1, 2, \dots$, where α_1 and α_2 are constants.

Solve the recurrence relation $a_n = 6a_{n-1} - 9a_{n-2}$ with $a_0 = 1$ and $a_1 = 6$.

Solution: The ch. eq. is $x^2 - 6x + 9 = 0 \Rightarrow (x - 3)^2 = 0$

$$\Rightarrow x_{1,2} = 3 \Rightarrow a_n = d_1 3^n + d_2 n \cdot 3^n$$

$$\begin{cases} 1 = d_1 \\ 6 = 3d_1 + 3d_2 \end{cases} \Rightarrow \begin{cases} d_1 = 1 \\ d_2 = 1 \end{cases} \Rightarrow \boxed{a_n = 3^n + n \cdot 3^n}$$

Answer Q4

Theorem 3

Suppose that the characteristic equation

$$x^{\textcircled{k}} - c_1 x^{k-1} - \dots - c_k = 0$$

has k distinct roots $x_1, x_2, \dots, x_k$.

Then a sequence $\{a_n\}$ is a solution of the recurrence relation

$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-\textcircled{k}}$ if and only if

$$a_n = \underline{\alpha_1} x_1^n + \underline{\alpha_2} x_2^n + \dots + \underline{\alpha_k} x_k^n$$

for $n = 0, 1, 2, \dots$, where $\alpha_1, \alpha_2, \dots, \alpha_k$ are constants.

Example

Find the solution to the recurrence relation

$$a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$$

with the initial conditions $a_0 = 2$, $a_1 = 5$, and $a_2 = 15$.

Solution: Ch. eq: $x^3 - 6x^2 + 11x - 6 = 0$ ($x=1$)

$$\Rightarrow x_1=1, x_2=2 \text{ and } x_3=3$$

$$a_n = d_1 \cdot 1^n + d_2 \cdot 2^n + d_3 \cdot 3^n$$

$$\begin{cases} 2 = d_1 + d_2 + d_3 \\ 5 = d_1 + 2d_2 + 3d_3 \\ 15 = d_1 + 4d_2 + 9d_3 \end{cases} \Rightarrow \begin{cases} d_1 = 1 \\ d_2 = 2 \\ d_3 = 2 \end{cases} \Rightarrow$$

$$a_n = 1 - 2^n + 2 \cdot 3^n$$

General k theorems

Theorem 4

Suppose that the characteristic equation

$$x^k - c_1x^{k-1} - \dots - c_k = 0$$

$$= (x-2)^2 (x-1)^5 x^6 (x-3)$$

has t distinct roots $x_1, x_2, \dots, x_t$ with multiplicities $m_1, m_2, \dots, m_t$, so that $m_1 + m_2 + \dots + m_t = k$.

Then a sequence $\{a_n\}$ is a solution of the recurrence relation

$a_n = c_1a_{n-1} + c_2a_{n-2} + \dots + c_ka_{n-k}$ if and only if

$$\begin{aligned} a_n &= (\alpha_{10} + \alpha_{11}n + \dots + \alpha_{1,m_1-1}n^{m_1-1})x_1^n \\ &\quad + (\alpha_{20} + \alpha_{21}n + \dots + \alpha_{2,m_2-1}n^{m_2-1})x_2^n \\ &\quad + \dots + (\alpha_{t0} + \alpha_{t1}n + \dots + \alpha_{t,m_t-1}n^{m_t-1})x_t^n \end{aligned}$$

for $n = 0, 1, 2, \dots$, where α_{ij} are constants.

Example

Suppose that the roots of the characteristic equation of a linear homogeneous recurrence relation are 2, 2, 2, 5, 5, and 9 (that is, there are three roots, the root 2 with multiplicity three, the root 5 with multiplicity two, and the root 9 with multiplicity one). What is the form of the general solution?

you do not have to find constants $d_1, d_2, \dots, d_k$

Solution: By Theorem 4, the general form of the solution is

$$a_n = (d_0 + d_1 n + d_2 n^2) 2^n + (d_3 + d_4 n) 5^n + d_5 9^n$$

Linear non-homogeneous recurrence relations

A **linear non-homogeneous recurrence relation degree k with constant coefficients** is a recurrence relation of the form

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k} + F(n),$$

where $c_1, c_2, \dots, c_k$ are real numbers, and $c_k \neq 0$.

A consequence of the principle of mathematical induction is that a sequence satisfying the recurrence relation in the definition is **uniquely determined** by this recurrence relation and the k initial conditions

$$a_0 = C_0, \quad a_1 = C_1, \dots, \quad a_{k-1} = C_{k-1}.$$

$$a_n = a_{n-1} + 2^n$$

$$f_n = f_{n-1} + f_{n-2} + n^2 + n + 1$$

$$a_n = a_{n-5} + n!$$

Theorem 5

If $\{a_n^{(p)}\}$ is a **particular solution** of the non-homogeneous linear recurrence relation with constant coefficients

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k} + \cancel{F(n)},$$

then every solution is of the form $\{a_n^{(p)} + a_n^{(h)}\}$, where $\{a_n^{(h)}\}$ is a solution of the associated homogeneous recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}.$$

Example

Solve the recurrence relation $a_n = 3a_{n-1} + 2n$ with $a_0 = 3$.

Solution: 1 step: solve homogeneous $a_n = 3a_{n-1}$,
ch. eq. $x - 3 = 0 \Rightarrow x = 3 \Rightarrow a_n^{(h)} = d_0 \cdot 3^n$

2 step: find particular solution: in
the form $cn + d$ $n^1: c = 3c + 2$

$$cn + d = 3(c(n-1) + d) + 2n \Rightarrow n^0: d = -3c + 3d$$

$$\Rightarrow c = -1, d = -3/2 \Rightarrow a_n^{(p)} = -n - \frac{3}{2}$$

$$\Rightarrow a_n = d_0 \cdot 3^n - n - \frac{3}{2}$$

$$a_0 = 3 \Rightarrow 3 = d_0 - \frac{3}{2} \Rightarrow d_0 = \frac{9}{2} \Rightarrow$$

$$\boxed{a_n = \frac{9}{2} 3^n - n - \frac{3}{2}}$$

Particular solutions theorem

Theorem 6

Suppose that $\{a_n\}$ satisfies the linear non-homogeneous recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k} + F(n),$$

and

$$F(n) = (b_t n^t + b_{t-1} n^{t-1} + \dots + b_1 n + b_0) \underline{s^n},$$

where $b_0, b_1, \dots, b_t$ and s are real numbers.

When s is not a root of the characteristic equation of the associated linear homogeneous recurrence relation, there is a particular solution of the form

$$(p_t n^t + p_{t-1} n^{t-1} + \dots + p_1 n + p_0) s^n$$

When s is a root of this characteristic equation and its multiplicity is m , there is a particular solution of the form

$$n^m (p_t n^t + p_{t-1} n^{t-1} + \dots + p_1 n + p_0) s^n.$$

Examples

What form does a particular solution of the linear nonhomogeneous recurrence relation

$$a_n = 6a_{n-1} - 9a_{n-2} + F(n)$$

have when $F(n) = 3^n$, $F(n) = n3^n$, $F(n) = n^2 2^n$, and $F(n) = (n^2 + 1)3^n$?

Solution: About homog. the root 3 is of mult 2 of ch. eq. of hom. rec. rel.

$$\text{If } F_n = 3^n \Rightarrow a_n^{(p)} = p_0 \cdot 3^n \cdot n^2$$

$$\text{If } F_n = n3^n \Rightarrow a_n^{(p)} = (p_0 + p_1 n) 3^n \cdot n^2$$

$$\text{If } F_n = n^2 \cdot 2^n \Rightarrow a_n^{(p)} = (p_0 + p_1 n + p_2 n^2) \cdot 2^n$$

$$\text{If } F_n = (n^2 + 1)3^n \Rightarrow a_n^{(p)} = (p_0 + p_1 n + p_2 n^2) \cdot 3^n \cdot n^2$$

Answer Q5

Homework:

- Read and understand EXAMPLE 3 at p. 505.
- Solve exercises # 7,11,13,19,21 at p. 510-511.
- Solve exercises # 1,3,9,13,15,19,21,23,27,29,31,33,37,45,47 at p. 524-526.
- Solve exercises # 1,3,7,9,12,13,17,19,21,35,37 at p. 535-536.